

UNIX COMMANDS WITH USAGES

APT-GET Command

sudo apt-get update

→ Update package lists.

sudo apt-get upgrade

→ Upgrade all upgradable packages.

sudo apt-get install [package]

→ Install a specified package.

sudo apt-get remove [package]

→ Remove a specified package.

sudo apt-get autoremove

→ Remove packages no longer needed.

AR Command

ar rcs lib.a file1.o file2.o

→ Create or replace a static library archive.

ar t lib.a

→ List contents of a static library archive.

ar x lib.a

→ Extract files from a static library archive.

ar d lib.a file.o

→ Delete a file from the archive.

ar q lib.a file.o

→ Quickly append file to archive.

ar s lib.a

→ Write an index to the archive.

CAT Command

cat file.txt

→ Display contents of a file.

cat file1.txt file2.txt

→ Concatenate and display multiple files.

cat > newfile.txt

→ Create a file by typing (end with Ctrl+D).

CD Command

cd /path/to/dir

→ Change to a specific directory.

cd ..

→ Go up one directory.

cd ~

→ Go to home directory.

CHMOD Command

chmod +x file.sh

→ Add execute permission.

chmod 755 file.sh

→ Set read/write/execute for owner, read/execute for others.

chmod -R 777 dir

→ Recursively set full permissions.

CHOWN Command

sudo chown user file

→ Change file owner.

sudo chown user:group file

→ Change file owner and group.

sudo chown -R user:group dir

→ Recursively change ownership.

CLEAR Command

clear

→ Clear the terminal screen.

CP Command

cp file1.txt file2.txt

→ Copy file1.txt to file2.txt.

cp -r dir1 dir2

→ Recursively copy directories.

cp -i file.txt backup/

→ Prompt before overwriting.

CRON Command

The cron table (crontab) defines time-based jobs that the system executes periodically (e.g., daily backups, weekly reports). It's essential for automating tasks in UNIX-like systems.

crontab -e

→ Edit current user's cron table (used to schedule recurring tasks and commands automatically).

crontab -l

→ List cron jobs.

crontab -r

→ Remove all cron jobs.

DF Command

df

→ Show disk space usage.

df -h

→ Show human-readable disk usage.

df -T

→ Show filesystem type.

DIFF Command

diff file1.txt file2.txt

→ Compare two files line by line.

diff -u old.c new.c

→ Unified format for patches.

DU Command

du

→ Show disk usage per file.

du -sh .

→ Summarize disk usage for current dir.

du -a

→ Show all files and folders with sizes.

EMACS Command

emacs file.txt

→ Open a file in Emacs.

emacs -nw file.txt

→ Open Emacs in terminal mode.

EXPORT Command

export VAR=value

→ Set environment variable.

export PATH=\$PATH:/new/path

→ Add to PATH.

FINGER Command

finger

→ Show info on all users.

finger username

→ Show info on specific user.

G++ Command

g++ file.cpp

→ Compile C++ file.

g++ -o prog file.cpp

→ Compile to a specific output file.

g++ -Wall file.cpp

→ Enable all warnings.

GCC Command

gcc file.c

→ Compile C file.

gcc -o prog file.c

→ Compile to output file.

gcc -Wall file.c

→ Enable all warnings.

gcc -fPIC -c file.c

→ Generate position-independent code.

GREP Command

grep 'main' file.c

→ Search for 'main' in file.

grep -i 'main' file.c

→ Case-insensitive search.

grep -r 'TODO' .

→ Recursively search all files.

GUNZIP Command

gunzip file.gz

→ Decompress .gz file.

GZIP Command

gzip file

→ Compress file.

gzip -k file

→ Compress and keep original.

HISTORY Command

history

→ Show command history.

history | grep ssh

→ Search history for 'ssh'.

KILL Command

kill PID

→ Send SIGTERM to a process.

kill -9 PID

→ Force kill a process.

kill -l

→ List signal names.

LAST Command

last

→ Show last logins.

last -x

→ Include shutdown/reboot entries.

LS Command

ls

→ List directory contents.

ls -l

→ Long listing format.

ls -a

→ Include hidden files.

ls -lh

→ Human-readable sizes.

LSCPU Command

lscpu

→ Display CPU architecture info.

MAKE Command

make

→ Run the default Makefile target.

make clean

→ Run the 'clean' target to delete built files.

make target

→ Run a specific Makefile target.

MAN Command

man ls

→ Show the manual for the ls command.

man 5 passwd

→ Show section 5 manual entry for passwd.

MKDIR Command

mkdir newdir

→ Create a new directory.

mkdir -p a/b/c

→ Create nested directories.

MORE Command

more file.txt

→ Display file contents one screen at a time, displays page by page.

MOUNT Command

sudo mount /dev/sdb1 /mnt

→ Mount a device to /mnt.

mount -t iso9660 -o ro /dev/cdrom /mnt/cdrom
→ Mount CD-ROM as read-only.

MV Command

mv old.txt new.txt
→ Rename a file.

mv file.txt /backup/
→ Move a file to another directory.

NSLOOKUP Command

nslookup example.com
→ Find the IP address of a domain.

PASSWD Command

passwd
→ Change your own password.

passwd username
→ Change another user's password (requires sudo).

PING Command

ping google.com
→ Ping a host continuously.

ping -c 4 example.com
→ Ping a host 4 times.

POPD Command

popd
→ Return to previous directory from stack.

PS Command

ps

→ Show current shell's processes.

ps aux

→ Show all running processes.

ps -ef

→ Show full-format process list.

PUSHD Command

pushd /tmp

→ Push current dir to stack and change to /tmp.

PWD Command

pwd

→ Print the current working directory.

RM Command

rm file.txt

→ Remove a file.

rm -r folder

→ Recursively remove a directory.

rm -f file.txt

→ Force delete a file without prompting.

RPM Command

rpm -i package.rpm

→ Install an RPM package.

rpm -q package

→ Query if a package is installed.

SFTP Command

sftp user@host

→ Start a secure FTP session.

put file.txt

→ Upload file to remote server.

get file.txt

→ Download file from remote server.

mkdir folder

→ Create a folder on remote.

cd /path

→ Change remote directory.

lcd /local/path

→ Change local directory.

ls

→ List remote directory contents.

pwd

→ Show remote working directory.

exit

→ Exit SFTP session.

SOURCE Command

source ~/.bashrc

→ Apply shell configuration changes without restarting.

. env.sh

→ Alternative syntax to source a file.

SSH Command

ssh user@host

→ Open a remote shell.

ssh -i key.pem user@host

→ Connect with an SSH key.

ssh -p 2222 user@host

→ Connect to SSH server on port 2222.

ssh user@host 'ls -la'

→ Run command remotely.

ssh -X user@host

→ Enable X11 forwarding.

SU Command (Switch User)

su john

→ Switch to the user john. This command allows you to assume another user's identity (often used for administrative access).

su -

→ Start a login shell as the root user. This loads the target user's environment variables as well.

su - username

→ Switch to a specific user with their full environment.

TAR Command

tar -cvf archive.tar files

→ Create a tar archive.

tar -xvf archive.tar

→ Extract from tar archive.

tar -xvzf archive.tar.gz

→ Extract gzipped archive.

tar -tvf archive.tar

→ List contents of tar without extracting.

tar -czf archive.tar.gz dir

→ Create gzipped archive from directory.

tar -xvjf archive.tar.bz2

→ Extract bzip2 compressed archive.

tar -cvJf archive.tar.xz dir

→ Create xz compressed archive.

TEE Command

echo 'hello' | tee file.txt

→ Write to file and display output.

ls | tee list.txt

→ Save and display ls output.

command | tee -a file.txt

→ Append output to file and display.

SCRIPT (TYPESCRIPT) Command

script

→ Start recording terminal session.

exit

→ Stop recording.

VI Command

vi file.txt

→ Edit file with vi.

vi +42 file.txt

→ Open file at line 42.

W Command

w

→ Show who is logged in and what they are doing.

WC Command

wc file.txt

→ Show line, word, byte counts.

wc -l file.txt

→ Show line count only.

WGET Command

wget http://example.com/file.txt

→ Download file from URL.

wget -O output.txt URL

→ Save downloaded file with a specific name.

WHO Command

who

→ Show who is logged in.

WHOAMI Command

whoami

→ Show current username.

YUM Command

yum install package

→ Install a package using yum.

yum update

→ Update all packages using yum.